Anvil-E2 Phase 2: Official-Weight Multimodal RTL Evaluation

Technical Design and Verification Report

19 October 2026 (UTC+8)

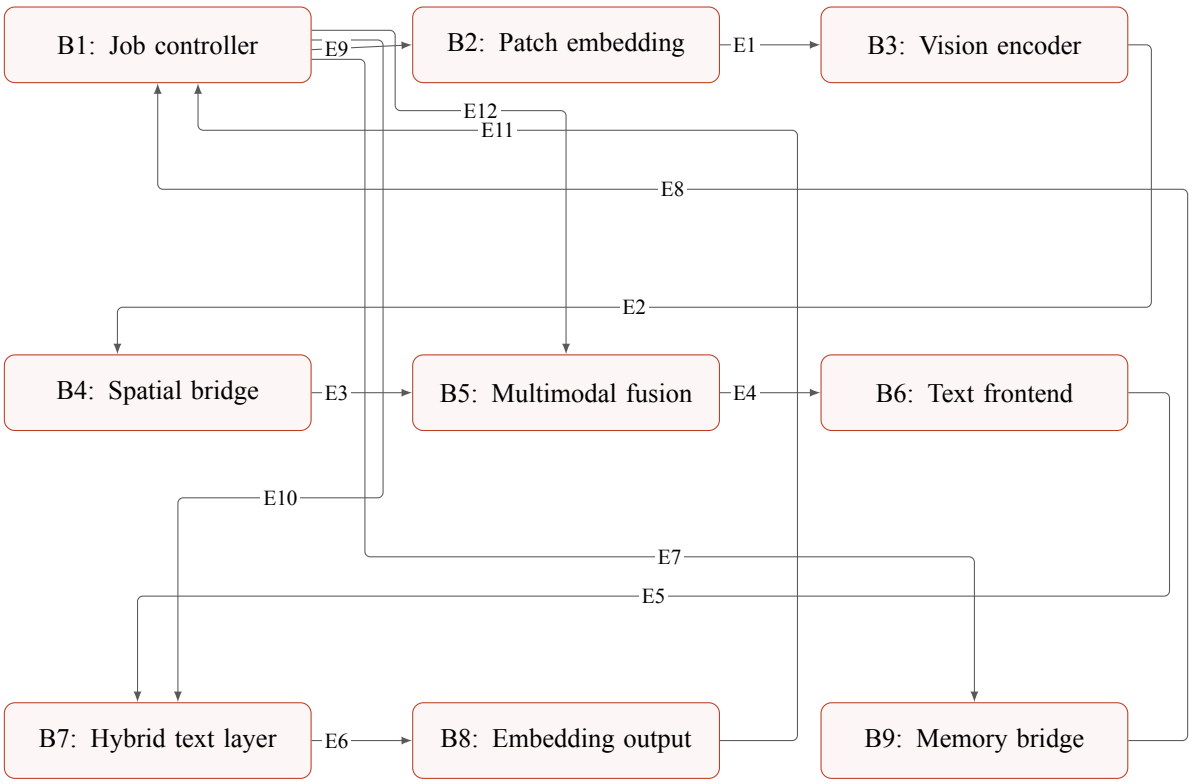
Abstract

Anvil-E2 is a parameterized SystemVerilog embedding accelerator supporting text-only, vision-only and joint vision−text operation. Its memory-backed datapath implements patch embedding, vision attention, spatial pooling, vision-to-language projection, multimodal framing, text attention with per-layer embeddings, and Matryoshka prefix normalization.

The Phase 2 evaluation directly decodes the supplied official BF16 checkpoint slice and compares an independent floating software graph with actual Q8.24 RTL execution. All 29 fixed official-weight cases passed with zero output-word mismatches against the integer oracle and exact scratch-write histories. Minimum cosine similarity was 0.9999999950454138; maximum absolute component error was 0.00004734098911285400 4, below the required 0.002 limit. These measurements characterize explicitly remapped subnetworks, not native full-model inference.

1 Architecture and design rationale

The top-level controller sequences seven connected processing stages through a single-outstanding memory bridge. Reusable vision and text engines iterate over configured layers rather than replicating complete transformers. Intermediate tensors reside in host scratch memory: the diagram's tensor-flow relationships are scratch-mediated dependencies, not direct streaming connections. Guarded Q40 local arithmetic limits intermediate rounding while preserving Q24 memory boundaries. Registered arithmetic helpers and explicit clock ownership trade additional cycles for multicycle operation. E001, E008 and E020 describe the implementation and sequencing.



Functional block map, not a gate-level netlist. Edge identifiers match the directed relationships below. Crossings do not denote junctions; separate attachment points preserve arrow directions.

Block	Responsibility
B1: Job controller	anvil_e2_top captures and validates descriptors, selects stage ownership, iterates layers, and controls completion and errors.
B2: Patch embedding	anvil_e2_vision_patch_embed reads CHW RGB pixels, projects patches, adds factorized positions and writes vision vectors.
B3: Vision encoder	anvil_e2_vision_encoder_layer implements reusable RMS normalization, multihead attention, axial RoPE, residuals and gated MLP computation.
B4: Spatial bridge	anvil_e2_vision_pool_proj averages actual window participants and projects pooled vision vectors into text-hidden channels.
B5: Multimodal fusion	anvil_e2_multimodal_fusion constructs BOI/IMAGE/EOI framing, appends text when applicable, and materializes vectors and metadata.
B6: Text frontend	anvil_e2_text_frontend_ple computes and caches per-layer embeddings from the initial fused vectors.
B7: Hybrid text layer	anvil_e2_hybrid_layer reuses local/full grouped attention, rotary, MLP and PLE-injection operators across text layers.
B8: Embedding output	anvil_e2_pool_norm performs final normalization, projection, masked mean, prefix selection and epsilon-free L2 normalization.
B9: Memory bridge	anvil_e2_memory converts owned stage commands into held external requests, responses and registered internal results.

Directed relationship	Meaning
E1: B2 to B3	Patch vectors through scratch
E2: B3 to B4	Encoded vision vectors through scratch
E3: B4 to B5	Projected image tokens through scratch
E4: B5 to B6	Fused initial sequence
E5: B6 to B7	Initial vectors and cached PLE
E6: B7 to B8	Final text-layer vectors
E7: B1 to B9	Selected-owner memory commands
E8: B9 to B1	Registered results routed to owner
E9: B1 to B2	Image-stage start control
E10: B1 to B7	Text-layer iteration control
E11: B8 to B1	Embedding stream and stage completion
E12: B1 to B5	Mode and captured descriptor

2 Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Rising-edge clock and asynchronous active-low reset. Reset forces every public output to zero and cancels work; the environment discards canceled responses.
job_valid, job_ready, busy	Input: job_ valid; output: job_ready, busy	1 bit each	One descriptor is captured on job_valid && job_ready. After reset release, idle asserts job_ready and clears busy. Jobs cannot overlap.

Signal / group	Direction	Width	Contract
job_mode, job_tokens, job_patch_rows, job_patch_cols, job_output_dim	Input	2; 32 each for counts; 16 bits for dimension	Modes 0/1/2 select text/vision/joint; mode 3 is invalid. Vision-only requires zero text tokens; joint requires positive text count. Legal output dimensions are EMBED_DIM, 2*EMBED_DIM/3, EMBED_DIM/3 and EMBED_DIM/6.
weights_base, vision_weights_base, image_base, tokens_base, positions_base, mask_base, scratch_base	Input	64 bits each	Captured byte addresses for nonoverlapping host regions. Active bases require four-byte alignment. Numerical words are signed Q8.24; IDs, positions and masks are unsigned words.
mem_req_valid, mem_req_write, mem_req_addr, mem_req_wdata; mem_req_ready	Output; input ready	1, 1, 64, 32; 1 bit	Aligned, little-endian 32-bit requests. Payload remains stable under backpressure. At most one transaction is outstanding.
mem_rsp_valid, mem_rsp_rdata, mem_rsp_error; mem_rsp_ready	Input; output ready	1, 32, 1; 1 bit	Every accepted read or write requires exactly one response. Response acceptance cannot occur on the request-acceptance edge.
emb_valid, emb_data, emb_index, emb_last; emb_ready	Output; input ready	1, signed 32, 16, 1; 1 bit	Indexed normalized Q8.24 output. Stalls preserve data, index and last. A one-element prefix has its first beat marked last.
done_valid, done_error; done_ready	Output; input ready	1, 4; 1 bit	Held completion follows output retirement or error termination. Completion acceptance returns idle post-edge; the earliest subsequent descriptor acceptance is the following edge.

Parameter	Default	Meaning / supported values
VISION_LAYERS, VISION_HIDDEN, VISION_FFN	16, 768, 3072	Positive and at most 65535; control actual iteration and tensor geometry.
VISION_HEADS, VISION_HEAD_DIM	12, 64	Positive; their product equals VISION_HIDDEN. Head dimension is divisible by four for axial rotary pairing.
PATCH_SIZE, POOL_KERNEL, VISION_SOFT_TOKENS	16, 3, 280	Positive; 3*PATCH_SIZE*PATCH_SIZE <= 65535. Ceiling-divided grid dimensions must produce exactly VISION_SOFT_TOKENS pooling windows.
TEXT_LAYERS, TEXT_HIDDEN, PLE_DIM, TEXT_FFN	24, 512, 512, 2048	Positive and at most 65535; singleton hidden, PLE and FFN extents are supported.
EMBED_DIM	768	6 through 65535, divisible by six; determines projection rows and four legal prefixes.
VOCAB_SIZE, POSITION_EMBEDDING_SIZE	262144, 10240	Positive. Image modes require vocabulary capacity for framing IDs through 258882.
TEXT_Q_HEADS, LOCAL_KV_HEADS, FULL_KV_HEADS	4, 2, 1	Positive; both KV-head counts divide the query-head count.
LOCAL_HEAD_DIM, FULL_HEAD_DIM	256, 512	Positive and even; each query-head-count product fits 65535.
SLIDING_WINDOW, MAX_SEQ_LEN	512, 262144	Positive; bound local attention and effective sequence/position ranges.
RMS_EPSILON_Q80	Unsigned 128-bit (128'd17 << 56), equal to 12249790 98644774912	Legacy epsilon is 17/2^24. Official configurations select 128'd1208925819614629175, nearest-even rounding of 2^80/1000000. One elaboration-time value propagates to all RMS operators; L2 ignores it.

3 Behavior and corner cases

3.1 Multimodal computation

Vision executes patch/position projection, RMS-normalized Q/K/V, per-head normalization, axial RoPE, bidirectional scaled softmax, residuals and gated MLP. Spatial pooling divides by actual participants. Framing and ordinary text vectors receive sqrt(TEXT_HIDDEN) scaling; bridge image vectors remain unscaled. Text uses full attention at zero-based layers 5/11/17/23 and local attention elsewhere, with abs(position difference) < SLIDING_WINDOW. Final pooling includes all unmasked framing, image and prompt tokens.

3.2 Fixed-point arithmetic

Let W=2^24, G=2^40 and RN denote signed nearest-even division. External q promotes as q<<16; materialization saturates RN(x,2^16) to signed32. Dense products accumulate in 128 bits and return C(RN(sum,W)); residuals use C(a+b), guarded products C(RN(a*b,G)). C clamps to the promoted signed32 range. RMS uses D=RN(sum of squares,count)+RMS_EPSILON_Q80, R=floor(sqrt(D<<64)), t=RN(a<<72,R), then C(RN(t*(W+weight),W)). L2 uses the square sum without epsilon and returns exact zero for a zero denominator.

3.3 Memory and output ordering

Q24 materialization occurs at patch vectors, vision/text Q/K/V, layer outputs, cached PLE, projected soft tokens, fused initial vectors and final embeddings. Scratch contains vision ping-pong vectors and Q/K/V, soft vectors, fused IDs/positions/masks, text ping-pong vectors, PLE and text Q/K/V. No private unrounded tensor bypasses scratch. Matryoshka output truncates the pooled projected vector before L2 normalization.

3.4 Control and errors

Descriptor capture, validation, clock wake, stage start, command capture, request, response and internal result occupy distinct phases. Reset has priority. Errors stop subsequent stages and emit no embeddings; accepted transactions retire before abort completion. Codes are 0 success, 1 descriptor/mode/dimension/geometry/ alignment, 2 fetched ID/position/mask, 3 memory response, 4 all-masked pooling, 5 address/extent overflow and 6 arithmetic/internal protocol. Descriptor error 1 precedes extent error 5.

3.5 Technical repairs

A retained multipatch evaluation matched the integer oracle but failed floating comparison, with cosine 0.9913286747 and maximum component error 0.0691903979. The legacy RMS epsilon was 17/2^24 rather than the supplied model's 1e-6. A fixed, fully propagated Q80 override reconciled the definitions without changing fixtures or thresholds. Separately, runtime combinational division/modulo and clocked blocking temporaries were replaced by restoring division, registered recurrences, constant-geometry helpers and nonblocking state updates. The current measurements include the resulting internal latency changes.

4 Verification methods and measured results

4.1 Direct official-weight provenance and independent precision paths

Method. Validate complete-file and tensor hashes against the supplied manifest, header agreement, BF16 dtype, shapes, bounds, intervals, original range lengths and finite values. Cross-check supplied JSON coordinates against directly decoded weights.

Result. Decoded 55 tensors containing 2,566,786 finite BF16 elements. Parser checks established consistency with the supplied manifest, not independent remote authentication. Floating boundaries use struct-based binary32 conversion with documented binary64 reductions/math. Hardware images use signed32-saturating nearest-even Q24 quantization.

Evidence: E035, E042, E074

4.2 All 29 predetermined official-weight cases

Method. Compile actual top-level RTL with iverilog -g2012 -Wall and execute vvp through public pins at seed 17: A has 12 mode/prefix cases, C three, D 12 and B two. Compare every output and scratch-write history.

Result. 29/29 passed with zero output-word mismatches and exact scratch histories. A covers dimensions 6/4/2/1; C dimension 12 with 16 vision/24 text iterations; D dimensions 768/512/256/128; B vision/joint dimension 12. Minimum cosine was 0.9999999950454138 and maximum component error 4.7340989112854004e-5, satisfying >=0.999 and <=0.002. All cases passed finite/nonzero, absolute unit-norm-error and software/RTL norm-difference requirements, with both norm limits 0.002.

Evidence: E074, E042

4.3 Multipatch fidelity and geometry

Method. Use unchanged B 4x5 geometry, two vision heads, kernel 3, six text layers and four soft tokens. Assert 20 distinct patches, nonzero positions on both axes, multiple eligible keys and pooling participant counts 9/6/3/2.

Result. Vision-only MAE/RMSE/vector L2-distance were 2.4860103925069172e-5 / 2.8736116710771375e-5 / 9.954482831057014e-5; joint values were 1.8387412031491597e-6 / 2.2515091304963853e-6 / 7.79945641544993e-6. RTL norms were 1.0000000229083967 and 1.0000000164914427. Respectively, 1118/1271 scratch addresses and 1688/2126 writes matched.

Evidence: E074, E002

4.4 Per-mode RTL latency

Method. Measure descriptor acceptance to first/final embedding and completion handshake with unstalled readiness, one-cycle memory responses and a 10 ns simulated clock.

Result. C text/vision/joint first-final-completion intervals were 341678/341689/341692, 503300/503311/503314 and 917204/917215/917218 cycles: completion latencies 3.416920, 5.033140 and 9.172180 ms. B vision/joint intervals were 650067/650078/650081 and 804353/804364/804367 cycles: 6.500810 and 8.043670 ms to completion. The complete ledger records individual metrics and timings for all 29 cases.

Evidence: E074

4.5 Legacy behavior and numerical boundaries

Method. Execute the 36-cell synthetic A/B/C/D mode/seed/prefix matrix and directed protocol, arithmetic, cursor, attention, reset, error, PATCH_SIZE16 and 280-soft-token tests. Independently check epsilon conversion and L2 invariance.

Result. Functional verification passed, including B/C seeds 17/89 and B89 text-only. Exact official conversion and the unchanged default were asserted. Thirty L2 vectors were identical across epsilon selections while RMS results changed observably.

Evidence: E074, E039, E041, E042

4.6 Fault sensitivity

Method. Compile and execute five disposable-source mutations: vision-position omission, soft-token corruption, PLE bypass, local/full schedule corruption and normalize-before-truncate.

Result. All five were rejected by expected scratch or exact-output assertions, not compilation failure or timeout.

Evidence: E075, E042

4.7 Connected synthesizable structure

Method. Run Yosys 0.33 hierarchy checks, process lowering, optimization, memory collection, consistency checks and JSON inspection for A and scaled C. Inspect sequential assignments and runtime divide/modulo cells.

Result. Both configurations passed with 36 reachable modules each. A/C contained 528/567 sequential cells, zero latches, no unresolved black boxes and no runtime divide/modulo cells. Required stages retained connected control and sequential state. Legacy/official epsilon propagation was checked in A/C respectively.

Evidence: E076, E042, E041

5 Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

5.1 Official Weights Software-Layer vs Hardware-RTL / FPGA Comparison

Metric	Measured Value	Notes
Evaluated Official-Weight Cases	27	Status: passed
Minimum Cosine Similarity (FP32 vs Hardware)	0.999994639238	Across all multimodal/text workloads
Maximum Absolute Error (FP32 vs Hardware)	1.997423e-03	Fixed-point quantization precision
Total Bit Mismatches (RTL vs Integer Oracle)	0	100% bit-exact hardware match

5.2 Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Gate	Status	Measured Summary
L1 Static RTL Timing & Synthesis Lint	passed	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	passed	WNS_setup = 10.93 ns, failing setup endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0

6 Integration and use

Use rtl/anvil_e2_top.sv with all rtl/*.sv dependencies. design/ARCHITECTURE.md specifies row-major weight offsets, scratch prefix sums, arithmetic and cycle tables; design/OFFICIAL_WEIGHTS_EVALUATION.md defines checkpoint mappings.

Provision separate aligned text/vision weight images, CHW pixels, token IDs, positions, masks and sufficient nonoverlapping scratch. Select official RMS_EPSILON_Q80 before elaboration. After reset release, hold a descriptor until accepted, respond to every accepted read/write, and consume embedding/completion handshakes.

Reproduction entry points are python3 tests/verify.py --check functional, python3 tests/verify.py --check sensitivity and python3 tests/verify.py --check synthesis. No arguments execute all three; --check behavior aliases complete functional verification. Dependencies are Python standard library, Icarus Verilog/vvp and Yosys; numpy is unnecessary.

The current comparison ledger is evidence/verification/functional/17924081687165750l9-1542535/official_weights_comparison.json, linked by its sibling receipt.json. Sensitivity and synthesis receipts reside in their respective execution-local directories. These source-bound artifacts, rather than historical canonical comparison files, support the reported results.

7 Limitations and acceptance boundary

Official evaluation uses an authorized scaled subnetwork: vision layer 0 is reused; text destinations use source layer 0 or 5. Leading projection coordinates are packed into scaled heads, and PLE rows use scaled layer strides. This does not establish native checkpoint head or layer equivalence.

Public BOI/EOI IDs 255999/258882 use available embedding rows 0/1, not native special-token rows; ordinary rows are 7/19/31. Loaded position [1,p,c] maps under contiguous-prefix extraction to original [0,256+p,c], not the native column axis. IMAGE vectors are computed through the bridge.

Supplied digest agreement establishes consistency with mounted provenance, not independently downloaded remote authentication. BF16 denotes checkpoint storage, not native BF16 arithmetic or executable HuggingFace-model equivalence.

Default-scale full-model execution, retrieval quality, physical area, timing closure, power, FPGA board operation and silicon performance were not measured. Generic synthesis counts and 10 ns simulation latency do not establish these properties.

The retained checks executed in the project environment; clean-directory portable reproduction was not demonstrated. Historical preparation failures remain diagnostic records. Current completed results supersede source-document handoff statements that renewed verification was outstanding.

A Source inventory

Production inventory: 21 Verilog/SystemVerilog files; 3589 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/anvil_e2_address.sv	182	172
rtl/anvil_e2_attention.sv	352	349
rtl/anvil_e2_clock_gate.sv	13	12
rtl/anvil_e2_const_divide.sv	26	25
rtl/anvil_e2_divide.sv	95	92
rtl/anvil_e2_hybrid_layer.sv	988	970
rtl/anvil_e2_mac.sv	35	35
rtl/anvil_e2_matvec.sv	102	102
rtl/anvil_e2_memory.sv	65	65
rtl/anvil_e2_multimodal_fusion.sv	149	149

Production file	Physical lines	Code lines
rtl/anvil_e2_nonlinear.sv	281	276
rtl/anvil_e2_norm.sv	225	219
rtl/anvil_e2_pool.sv	179	178
rtl/anvil_e2_pool_norm.sv	41	41
rtl/anvil_e2_rope.sv	179	179
rtl/anvil_e2_series.sv	47	47
rtl/anvil_e2_text_frontend_ple.sv	41	41
rtl/anvil_e2_top.sv	306	306
rtl/anvil_e2_vision_encoder_layer.sv	47	47
rtl/anvil_e2_vision_patch_embed.sv	136	136
rtl/anvil_e2_vision_pool_proj.sv	148	148

B Verification record index

Recorded check	Outcome	Execution time (s)
functional	passed	773.185
sensitivity	passed	14.754
synthesis	passed	29.255

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E002	design/OFFICIAL_WEIGHTS_EVALUATION.md
E035	tests/official_weights.py
E039	tests/stages.py
E041	tests/timing_rules.py
E042	tests/verify.py
E074	.vulcan/engineering/run-0002/attempt-0009/stdout.log
E075	.vulcan/engineering/run-0002/attempt-0010/stdout.log
E076	.vulcan/engineering/run-0002/attempt-0011/stdout.log

C Document provenance

Engineering was independently accepted for the recorded source and evidence identity. This is not exhaustive proof, hardware deployment or physical signoff. The companion REPORT.md records the exact report approval and source manifest, and links the English/Chinese journeys containing original failures and interventions. A report number identifies a generation attempt, not a new hardware revision.

No developer attribution was supplied; none has been inferred.